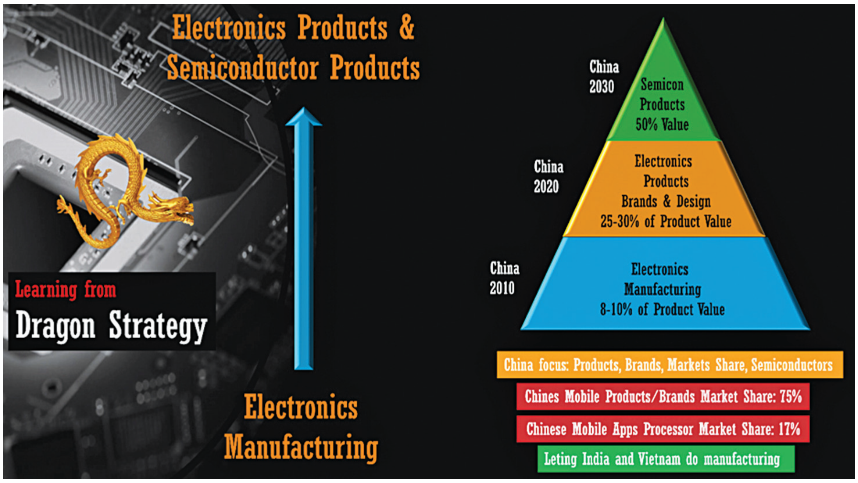

Fig. 3: Learning from Dragon Strategy

established players like Qualcomm, Samsung, and MediaTek soon.

In India, it is also important that we move from electronics manufacturing to electronics products and semiconductor chips. To chart a course similar to China's, Indian brands must focus on owning product designs and components. Reclaiming ownership of the electronics landscape is crucial for long-term success and self-reliance. By shifting from being a manufacturing hub to creating and owning semiconductor components and manufacturing, India can assert its presence on the global stage.

From challenges to opportunity

In the ever-evolving landscape of India's electronics industry, the intertwined and dependent entities of electronics products and semiconductor chips have come to the forefront, showcasing both challenges and opportunities. To better understand the situation, let's examine the broader Indian market for electronics products.

Only around 10% of these products originate from Indian brands and companies. Global brands dominate the remaining 90%. For long-term sustainability and bringing in-depth manufacturing as a next step, this ratio needs a better balance for India's self-reliance in electronics. On the semiconductor products front, more than 98% of the chips used in $180 billion of electronics products come from global companies and brands.

There is no doubt that in today's interdependent world economy, global products and brands play a very crucial role, striving for an increased market share, particularly in terms of Indian products, Indian brands, and Indian design, is a well-deserved aspiration. It is estimated that the electronics market will grow to $400 billion before the end of this decade. Capturing a well-deserved 33% of the market share in electronics products, up from the current less than 10%, could unlock a market potential of $133 billion. Similarly, the semiconductor market will grow to $100 billion, and the 33% market share for Indian chips will unlock a $33 billion economic opportunity.

These numbers underscore the potential for India to make significant strides in the global electronics and semiconductor landscape. A host of sectors eagerly await India's prowess in electronics and semiconductors. AI/ML, digital education, digital healthcare, fintech, telecom, transport, smart cities, data centres, and server domains rely heavily on these technologies.

Beyond market share and economic benefits, India's cultural ethos offers a unique advantage—a commitment to sustainability and durability. Unlike the 'use and throw' culture seen worldwide recently, Indians tend to value and extend the life of their products. This presents an opportunity to create 'reusable, re-pairable, and re-cyclable' electronics products that align with the principles of a circular economy and sustainable growth. Transitioning from an electronics manufacturing hub to an electronics product nation, a global centre for innovation, design, and product ownership is the key to India's success.

As we embark on this transformative journey, various domains beckon our attention. A few such bright spots for immense opportunities could be electric vehicles and drones. The automotive sector, for instance,

The Government's Initiatives

The Indian government has introduced several initiatives to bolster the semiconductor ecosystem.

- Under the India Semiconductor Mission, semiconductor manufacturing facilities receive incentives to cover up to 50% of their costs on a Pari-passu basis.
- Additional 20-25% incentives for semiconductor manufacturing by governments like Gujarat, Odisha, Uttar Pradesh, Tamil Nadu, Karnataka, and Telangana.
- 'Design Linked Incentives (DLI)' scheme also part of FutureDesign Initiative, which offers incentives to semiconductor product companies and startups involved in chip design. This scheme covers 50% of project costs and provides access to EDA tools and resources, making chip design more accessible and affordable.
- To address the talent gap, the government has implemented programmes such as 'Chips to System' (C2S), which aims to produce 175 ASICs, 30 SoC (system-on-chip) designs, and 30 IP (intellectual property) blocks and support more than 150 academic institutes.
- A new B.Tech degree programme in electronics and VLSI (very large-scale integration) design has been approved by the All India Council for Technical Education (AICTE). This degree will pave the way for a more extensive pool of electronics and VLSI design professionals. So far, more than 130 institutes have signed up to offer the B.Tech degree in Electronics & VLSI Design.
- ChipIn Centre at C-DAC and Future LABs initiatives to support all these initiatives.